

Do AI Data centers actually “use up” water?

Posted by u/sirepicness666 • 0 points • 59 comments

Now I'm not a supporter of them, but I hear a lot of people saying that they're “using up” fresh water, but wouldn't the water cycle recirculate the water back into the eco-system to be reused again?

Comments

u/lm_a_starfish • 1 points • 2026-03-24 14:39 UTC

From my understanding, the issue is that they are using freshwater, which is essential to humans, and their wastewater is not fresh water. so ya, they recirculate water but its not fresh water.

And it impacts local communities. So places where a bunch of data centers are clustered will be the ones that feel it. From what I read, dry air is very efficient for cooling, but since those places have less water, it hits the locals hard.

There are definitely ways that data centers can cool with less water or even no water, and it already exists and some data centers use that. I do hope they get implemented worldwide. and maybe they can figure out a way to use non fresh water by making new data centers able to withstand the “infected” water. Maybe even use it and somehow turn it into fresh water!

But thats extremely optimistic. This will all take research and money and well, will companies wanna do that?

u/SadEntertainer9808 • 1 points • 2026-03-24 02:06 UTC

Some water is lost to evaporation in swamp cooling, reentering the water cycle but being not immediately available for further use. For training and inference combined for the four big US labs, this quantity of water happens to be about the amount used by four average square miles of US farmland. There are 1,370,000 square miles of farmland in the US.

u/General_Exception • 1 points • 2026-03-23 23:32 UTC

Modern AI datacenters use evaporation to cool their servers.

More fresh water is evaporated PER DAY by Mother Nature from lakes, rivers, and ponds, than all the datacenters combined per year.

Agriculture “consumes” more water than datacenters.